Light

Legal, Ethical & Environmental Issues

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Edexcel GCSE Computer Science — **Legal, Ethical and Environmental Issues (1CP2/01 Topic 5: Issues and Impact)**. This topic covers the environmental impact of computing, legal protections for data and intellectual property, and ethical considerations around technology use including AI and automation.

Exam questions typically ask you to **describe** environmental impacts, **identify** methods of intellectual property protection, **explain** legal requirements for data handling, and **discuss** ethical implications of technology. The 2024 Paper 1 included questions on replacement cycles and environmental impact, and IP protection methods. Expect questions worth 2—6 marks requiring you to link issues to specific contexts.

Why this topic matters

Technology impacts society in many ways beyond just its technical function. As technology becomes more integrated into our lives, understanding these broader issues becomes increasingly important:

- **Environmental responsibility:** The tech industry has a significant carbon footprint and creates millions of tonnes of e-waste annually — understanding these impacts helps us make better choices

- **Legal compliance:** Organisations must follow laws protecting personal data and intellectual property — breaking these laws can result in significant fines and legal action
- **Ethical awareness:** As AI and automation become more prevalent, understanding their ethical implications is increasingly important for citizens and policymakers
- **Career relevance:** All technology professionals need to understand the broader impact of their work on society, the environment, and individuals
- **Consumer awareness:** Understanding data protection and privacy rights helps you protect your own information online
- **Exam relevance:** This topic appears in Paper 1 Question 5, typically worth 10—15 marks across several sub-questions

What you must know

This checklist covers the key exam-visible concepts from Edexcel 1CP2 Topic 5: Issues and Impact. You should be able to explain each point and apply it to exam scenarios.

- **Describe** the environmental impact of computing: energy consumption during use, resources used in manufacture, and problems of disposal
- **Explain** how replacement cycles affect environmental impact and how extending device lifespan reduces harm
- **Identify** methods of protecting intellectual property: copyright, patents, trademarks, and licensing
- **Describe** the legal requirements for handling personal data under the Data Protection Act
- **Explain** ethical issues around privacy, consent, and ownership of personal data
- **Discuss** ethical and legal issues related to AI, machine learning, and robotics including accountability, bias, and safety

> **Specification link:** This topic covers Edexcel 1CP2 Paper 1 Topic 5: Issues and Impact — specifically environmental issues (energy, manufacture, replacement cycles, disposal), ethical and legal issues around personal data (privacy, ownership, consent, misuse, Data Protection), ethical and legal issues for AI/ML/robotics (accountability, bias, safety, liability), and intellectual property protection (copyright, patents, trademarks, licensing).

Toolkit

These are the key patterns, definitions and relationships you need to apply in exam questions.

Environmental issues summary

Stage	Environmental issue	Key exam phrases
Manufacture	Extraction of raw materials, energy used in production	"Extraction of metals", "fossil fuels used", "water consumption"
Use	Energy consumption during operation	"Electricity consumption", "carbon footprint", "data centres"

Stage	Environmental issue	Key exam phrases
Disposal	E-waste, toxic materials, landfill	"E-waste", "hazardous materials", "landfill", "pollution"
Replacement cycle	Frequent upgrades increase all of the above	"Shorter replacement cycles", "more manufacturing", "more disposal"

Intellectual property protection

Protection type	What it protects	Example
Copyright	Creative works (text, music, images, software code)	A song, a photograph, a computer program
Patent	Inventions and new processes	A new type of processor, a hardware invention
Trademark	Brand names, logos, slogans	Company logo, product name, advertising slogan
Licensing	Grants permission to use protected work under specific terms	Software licence, music streaming rights

Data Protection Act principles

The Data Protection Act requires organisations to handle personal data according to key principles:

- **Lawfulness and fairness:** Data must be collected and used legally and fairly
- **Purpose limitation:** Data should only be used for the purpose it was collected
- **Data minimisation:** Only collect data that is necessary
- **Accuracy:** Data must be kept accurate and up to date
- **Storage limitation:** Data should not be kept longer than necessary
- **Security:** Data must be kept secure against unauthorised access

Ethical issues with AI/ML/Robotics

Issue	Description	Example
Accountability	Who is responsible when AI makes a mistake?	Self-driving car causes an accident — who is liable?
Bias	AI systems may reflect biases in training data	Facial recognition less accurate for certain demographics
Safety	Ensuring AI systems operate safely	Medical diagnosis AI must be reliable

Issue	Description	Example
Job displacement	Automation replacing human workers	Robots in factories, automated customer service
Privacy	AI systems processing personal data	Facial recognition in public spaces

1. Environmental issues

Computing technology has a significant environmental impact throughout its lifecycle — from manufacture, through use, to disposal. Understanding these impacts is essential for making informed decisions about technology use.

1.1 Manufacturing impact

The production of computers and electronic devices requires significant resources and energy:

Raw materials:

- **Rare earth metals:** Mining for materials like lithium, cobalt, and gold causes environmental damage
- **Water usage:** Manufacturing semiconductors and chips requires large amounts of water
- **Plastic and glass:** Cases, screens, and components use non-renewable resources

Energy consumption:

- Factories require significant electricity, often from fossil fuels
- Global supply chains mean materials and components are transported long distances
- The carbon footprint of manufacturing a single smartphone can be substantial

1.2 Energy use during operation

Electronic devices consume electricity throughout their operational life:

- **Personal devices:** Computers, phones, and tablets need regular charging
- **Data centres:** Servers that power the internet consume enormous amounts of electricity
- **Cooling systems:** Data centres require additional energy for cooling
- **Network infrastructure:** Routers, switches, and transmission equipment run continuously

If this electricity comes from fossil fuels, it contributes to carbon emissions and climate change.

> **Exam tip:** When discussing environmental impact, always consider all three stages: manufacture, use, and disposal. A complete answer addresses the full lifecycle.

1.3 Disposal and e-waste

When electronic devices reach the end of their life, they become **e-waste** (electronic waste). This creates several environmental problems:

Hazardous materials:

- **Lead:** Found in solder and CRT monitors
- **Mercury:** Used in some displays and switches
- **Cadmium:** Found in batteries and semiconductors
- **Brominated flame retardants:** In plastic casings

When e-waste is not properly recycled, these toxic materials can leak into soil and water, causing pollution and health risks.

Scale of the problem:

- Millions of tonnes of e-waste are generated globally each year
- Much e-waste is exported to developing countries with less strict regulations
- Only a fraction of e-waste is properly recycled

1.4 Replacement cycles

The **replacement cycle** refers to how often devices are replaced with newer models. Shorter replacement cycles increase environmental impact:

- **More manufacturing:** More frequent replacement means more devices need to be produced
- **More disposal:** Old devices become e-waste more quickly
- **More resource extraction:** Greater demand for raw materials

Factors driving short replacement cycles:

- New features and improved performance in newer models
- Software updates that may slow down older devices
- Fashion and marketing encouraging upgrades
- Batteries that degrade over time
- Devices designed to be difficult to repair

1.5 Reducing environmental impact

There are several ways to reduce the environmental impact of computing. Individuals, organisations, and manufacturers all have a role to play:

Extend device lifespan

- Keep devices longer before replacing
- Repair rather than replace
- Choose devices designed for longevity
- Upgrade components instead of whole devices

Responsible disposal

- Use certified e-waste recycling
- Donate working devices
- Return devices to manufacturers
- Never put electronics in general waste

Reduce energy use

- Turn off devices when not in use
- Use energy-efficient settings
- Choose renewable energy sources
- Select energy-efficient devices

1.6 The role of manufacturers

Manufacturers can help reduce environmental impact through responsible design and business practices:

Design for longevity:

- Make devices that last longer
- Design products that can be repaired

- Provide spare parts and repair guides
- Support software updates for longer periods

Design for recycling:

- Use fewer different materials
- Make devices easier to disassemble
- Avoid gluing components together
- Label materials clearly for recyclers

Responsible manufacturing:

- Use renewable energy in factories
- Source materials responsibly
- Reduce water consumption
- Minimise packaging waste

Take-back schemes:

- Accept old devices for recycling
- Offer trade-in programmes
- Ensure responsible processing of returned devices
- Recover valuable materials for reuse

> **Exam tip:** When asked how to reduce environmental impact, think about different stakeholders — individuals can extend device lifespan, organisations can recycle responsibly, and manufacturers can design for longevity and recycling.

1.7 Environmental impact diagram

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┌──────────────────────────────────────────────────────────────────────────────────────┐
│ DEVICE LIFECYCLE ENVIRONMENTAL IMPACT │
├──────────────────────────────────────────────────────────────────────────────────────┤
│ │ │ MANUFACTURE USE DISPOSAL │ │ ┌──────────┐ ┌──────────┐
┌──────────┐ │ │ │ • Mining │ │ • Elec- │ │ • E-waste │ │ │ │ metals │ │
tricity │ │ │ • Landfill │ │ │ │ • Water │──────►│ • Data │──────►│ •
Toxic │ │ │ │ use │ │ centres │ │ chemicals│ │ │ │ • Factory │ │ • Cooling │
│ • Pollution│ │ │ │ energy │ │ │ │ │ │ │ └──────────┘ └──────────┘
└──────────┘ │ │ │ │ │ │ │ REPLACEMENT CYCLE │ │ │
│◄──────────────────────────────────────────────────────────────────────────────────────┘ │ │ │ │ │ Shorter
replacement cycles = More manufacturing + More disposal │ │ Longer
replacement cycles = Less environmental impact │ │ │
└──────────────────────────────────────────────────────────────────────────────────────┘
```

Worked Example 1 – Describing manufacturing impact

Question: Describe one environmental impact of manufacturing electronic devices. [2 marks]

Model answer:
Manufacturing requires the extraction of rare earth metals through mining **(1)**. This mining process damages the environment / uses large amounts of energy / creates pollution **(1)**.

Exam tip: For "describe" questions, state the issue and then expand on its effect. Link your two points together for a complete answer.

Worked Example 2 – Explaining e-waste problems

Question: Explain why e-waste is an environmental problem. [2 marks]

Model answer:
Electronic devices contain hazardous materials such as lead and mercury **(1)**. When disposed of improperly, these toxic materials can leak into soil and water, causing pollution **(1)**.

Exam tip: Name specific hazardous materials (lead, mercury, cadmium) rather than just saying "chemicals" — this shows better understanding.

Worked Example 3 – Explaining replacement cycles

Question: Explain how shorter replacement cycles affect the environment. [2 marks]

Model answer:
Shorter replacement cycles mean devices are disposed of more frequently **(1)**. This leads to more e-waste / more landfill / more pollution **(1)**.
Alternative answer:
Shorter replacement cycles mean more devices need to be manufactured **(1)**. This increases extraction of metals / increased use of water / more fossil fuels used **(1)**.

Exam tip: This question appeared in the 2024 exam. You can approach it from either the disposal angle OR the manufacturing angle — both are valid.

Worked Example 4 – Describing energy consumption

Question: Describe how data centres contribute to environmental impact. [2 marks]

Model answer:
Data centres consume large amounts of electricity to power servers **(1)**.
They also require additional energy for cooling systems to prevent overheating **(1)**.

> **Exam tip:** Data centres are a common exam context. Remember they need power for both running servers AND cooling them.

2. Legal issues

Technology raises important legal questions around protecting intellectual property and personal data. Understanding these legal frameworks is essential for anyone working with technology.

2.1 Intellectual property (IP)

Intellectual property refers to creations of the mind — inventions, creative works, designs, and brand identities. The law provides several ways to protect IP:

Copyright

Copyright automatically protects original creative works, including:

- Written text (books, articles, websites)
- Music and sound recordings
- Images, photographs, and artwork
- Films and videos
- Computer software and code

Copyright gives the creator exclusive rights to copy, distribute, and adapt their work. Others cannot use copyrighted material without permission.

Patents

A **patent** protects inventions — new products or processes that provide a technical solution. In computing, patents might protect:

- Hardware inventions (new processor designs)
- Manufacturing processes
- Technical methods and algorithms (in some jurisdictions)

Patents must be applied for and granted — they do not arise automatically like copyright.

> **Exam tip:** Know the difference: Copyright protects creative works and is automatic. Patents protect inventions and must be applied for.

Trademarks

A **trademark** protects brand identity — the signs that distinguish one business's goods or services from another:

- Company names and logos
- Product names
- Slogans and taglines
- Distinctive colours or sounds associated with a brand

Trademarks prevent others from using similar branding that could confuse customers.

Licensing

Licensing is a way for IP owners to grant others permission to use their protected work under specific terms:

- **Software licences:** Define how software can be used (single user, site licence, etc.)
- **Open source licences:** Allow free use with certain conditions

- **Creative Commons:** Various levels of permission for creative works

A licence is a legal agreement — using software or content outside the licence terms is illegal.

2.2 Why IP protection matters

For creators:

- Ensures creators can benefit financially from their work
- Encourages innovation by protecting investment in R&D
- Prevents others from unfairly profiting from someone else's work

Consequences of infringement:

- Legal action and financial penalties
- Requirement to pay damages to the rights holder
- Criminal prosecution for serious piracy

2.3 Software piracy

Software piracy is the illegal copying, distribution, or use of software without proper licensing. This is a form of copyright infringement.

Examples of software piracy:

- Installing software on more computers than the licence allows
- Downloading software from illegal file-sharing sites
- Using software without purchasing a valid licence
- Sharing software with others without permission

Why piracy is harmful:

- Developers lose income they need to create and maintain software
- Reduces funding for innovation and new features
- Pirated software may contain malware
- Users miss out on updates and support

2.4 IP protection matching diagram

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┌──────────────────────────────────────────────────────────────────────────────────┐
│ INTELLECTUAL PROPERTY PROTECTION │
├──────────────────────────────────────────────────────────────────────────────────┤
│ │ │ WHAT IS BEING PROTECTED? USE THIS PROTECTION: │ │ │ │
┌──────────────────────────┐ ┌──────────────────┐ │ │ │ Creative work
│──────────────►│ COPYRIGHT │ │ │ │ (music, text, images, │ │ (automatic) │ │ │
software code) │ └──────────────────┘ │ │ └──────────────────────────┘ │
│ │ ┌──────────────────────────┐ ┌──────────────────────┐ │ │ │ Invention
│──────────────►│ PATENT │ │ │ │ (hardware, new process) │ │ (must apply) │ │ │
└──────────────────────────┘ └──────────────────────┘ │ │ │
┌──────────────────────────┐ ┌──────────────────┐ │ │ Brand identity
│──────────────►│ TRADEMARK │ │ │ (logo, name, slogan) │ │ (register) │ │ │
└──────────────────────────┘ └──────────────────┘ │ │ │
└──────────────────────────────────────────────────────────────────────────────────┘
```

2.5 Data Protection Act

The **Data Protection Act** is UK legislation that controls how personal data is collected, stored, and used. It gives individuals rights over their data and places obligations on organisations.

What counts as personal data?

Any information that can identify a living individual:

- Names and addresses

- Email addresses and phone numbers
- Date of birth
- Financial information
- Medical records
- Photos and videos of identifiable people
- IP addresses and online identifiers

Key requirements for organisations

- Only collect data that is necessary
- Tell people why data is being collected
- Keep data accurate and up to date
- Keep data secure
- Only keep data as long as needed
- Not share data without permission
- Allow individuals to access their data
- Allow individuals to request deletion

Individual rights under Data Protection

The Data Protection Act gives individuals important rights over their personal data:

Rights relating to access

- **Right to be informed:** Know what data is held and why
- **Right of access:** Request a copy of your data
- **Right to rectification:** Have incorrect data corrected

Rights relating to control

- **Right to erasure:** Request deletion of data
- **Right to object:** Stop certain processing
- **Right to data portability:** Transfer data to another provider

Worked Example 5 — Identifying IP protection

Question: A company has developed a new hardware invention. State the appropriate method of intellectual property protection. [1 mark]

Model answer:
Patent **(1)**

Exam tip: This question type appeared in the 2024 exam. Learn the simple matching: inventions = patent, creative works = copyright, brand identity = trademark.

Worked Example 6 – Matching IP protection types

Question: Complete the table to show the appropriate protection for each type of intellectual property. [2 marks]

Intellectual property	Method of protection
A hardware invention	_____
An advertising slogan	_____

Model answer:
A hardware invention → **Patent (1)**
An advertising slogan → **Trademark (1)**

Exam tip: Slogans and logos are trademarks (brand identity). Software code is copyright (creative work). Hardware inventions are patents.

Worked Example 7 – Explaining copyright

Question: A musician has released a new song. Explain how copyright protects the musician. [2 marks]

Model answer:
Copyright gives the musician exclusive rights to copy and distribute their song **(1)**.
Others cannot use the song without permission, allowing the musician to earn money from their work **(1)**.

Exam tip: Always link the protection to the benefit — copyright protects the work AND allows the creator to benefit financially.

Worked Example 8 – Describing Data Protection requirements

Question: A school stores personal data about students. State two requirements of the Data Protection Act that the school must follow. [2 marks]

Model answer:
The school must keep the data secure / prevent unauthorised access **(1)**.
The school must only keep the data for as long as necessary **(1)**.

> **Exam tip:** Learn several Data Protection principles so you can choose different ones for different questions. Common ones: security, accuracy, purpose limitation, data minimisation.

3. Ethical issues

Beyond legal requirements, technology raises ethical questions about right and wrong. These issues are particularly important as AI, machine learning, and robotics become more prevalent.

3.1 Privacy and consent

Technology has made it easier than ever to collect and analyse personal data, raising important ethical questions:

Privacy concerns:

- How much data collection is acceptable?
- Should people be tracked online or in public spaces?
- Who should have access to personal data?
- What happens to data after it is collected?

Consent issues:

- Do people truly understand what they are consenting to?
- Are privacy policies readable and clear?
- Is it ethical to require consent as a condition of using a service?
- Can children meaningfully consent to data collection?

3.2 Ownership of data

Who owns the data that is collected about individuals? This is a complex ethical and legal question:

- Does a social media company own posts made by users?
- Who owns data generated by fitness trackers or smart devices?
- Should individuals be able to sell their own data?
- What rights do people have over their digital footprint?

Data as a commodity:

Personal data has become extremely valuable to businesses. Companies collect data to:

- Target advertising more effectively
- Develop new products based on user behaviour
- Sell to third parties for marketing
- Train AI and machine learning systems

This raises ethical questions about whether individuals should benefit from the value of their own data, and whether the current model of "free services in exchange for data" is fair.

> **Exam tip:** Ethical questions often do not have a single "right" answer. In exam responses, show you understand multiple perspectives.

3.3 AI, machine learning, and robotics

As AI systems make more decisions that affect people's lives, several ethical issues arise:

Accountability

When AI systems make mistakes, who is responsible?

- If a self-driving car causes an accident, who is liable — the manufacturer, the programmer, or the owner?
- If an AI makes a wrong medical diagnosis, who is accountable?
- How can victims seek redress when harmed by AI decisions?

Bias in AI systems

AI systems learn from data, and if that data contains biases, the AI will reproduce them:

- Facial recognition may be less accurate for certain demographic groups
- Recruitment AI trained on historical data may discriminate
- Credit scoring algorithms may disadvantage certain communities

This raises questions about fairness and discrimination in automated decision-making.

Safety

AI and robotics systems must operate safely, especially in critical applications:

- Medical diagnosis systems must be reliable
- Autonomous vehicles must not endanger lives
- Industrial robots must protect workers

Job displacement

Automation and AI may replace human workers:

- Manufacturing robots replacing factory workers
- AI customer service replacing call centre staff
- Automated systems replacing administrative roles

This raises questions about economic impact and responsibility for displaced workers.

> **Beyond the spec / For interest:** New technologies continue to raise ethical questions, such as autonomous vehicles (liability in accidents), virtual assistants (always-listening devices), and AI-generated content (misinformation). These examples illustrate how the core ethical issues of accountability, bias, safety, and privacy apply to emerging technologies.

3.4 Ethical issues diagram

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┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│ ETHICAL ISSUES IN COMPUTING │
├──────────────────────────────────────────────────────────────────────────────────────────────────────────────┤
│ │ │ PERSONAL DATA AI / ML / ROBOTICS │ │ ┌──────────────────────┐
┌──────────────────────┐ │ │ │ Privacy │ │ Accountability │ │ │ │ • How much │
│ • Who is │ │ │ │ collection is OK?│ │ responsible? │ │ │ │ │ │ │ │ │ │
Consent │ │ Bias │ │ │ │ • Do people truly │ │ • AI reflects │ │ │ │
understand? │ │ training data │ │ │ │ │ │ │ │ │ Ownership │ │ Safety │ │ │
│ • Who owns user │ │ • Must operate │ │ │ │ data? │ │ reliably │ │ │ │ │ │
│ │ │ │ │ Job displacement │ │ │ │ │ │ • Automation │ │ │ │ │ │ replacing
workers│ │ │ └──────────────────────┘ └──────────────────────┘ │ │ │ │ Key
question: Just because we CAN do something, does that mean we │ │ SHOULD do
it? │ │ │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
```

3.5 Balancing benefits and risks

Technology often involves trade-offs between benefits and ethical concerns. Responsible use of technology requires considering both sides:

Example: Facial recognition

Benefits:

- Can help find missing persons
- Can identify criminals and terrorists
- Convenient for security access

Concerns:

- Mass surveillance without consent
- Bias leading to wrongful identification
- Erosion of privacy in public spaces

Example: Personalised advertising

Benefits:

- More relevant ads for consumers
- Funds free online services
- Helps small businesses reach customers

Concerns:

- Extensive tracking of online behaviour
- Manipulation of consumer behaviour
- Filter bubbles limiting information

3.6 Making ethical decisions

When faced with ethical decisions about technology, individuals and organisations should consider:

Questions to ask:

- Who benefits from this technology?
- Who might be harmed?
- Is consent truly informed?
- Are there alternatives with fewer ethical concerns?
- What would happen if everyone did this?

Stakeholders to consider:

- Users and consumers
- Employees and workers
- Shareholders and investors
- The wider community
- The environment
- Future generations

> **Exam tip:** When answering "discuss" or "evaluate" questions, consider multiple stakeholders and their different perspectives. A good answer acknowledges that ethical issues rarely have simple solutions.

Worked Example 9 – Explaining AI accountability

Question: A self-driving car is involved in an accident. Explain one ethical issue this raises. [2 marks]

Model answer:
It raises the issue of accountability — it is unclear who is responsible for the accident **(1)**.
It could be the car manufacturer, the software developer, or the owner, making it difficult for victims to seek compensation **(1)**.

Exam tip: Accountability is a key ethical issue for AI. Always consider who is responsible when things go wrong with automated systems.

Worked Example 10 – Describing AI bias

Question: Explain why bias in AI systems is an ethical concern. [3 marks]

Model answer:
AI systems learn from training data, and if this data contains biases, the AI will reproduce them **(1)**.
This could lead to unfair treatment of certain groups, for example facial recognition being less accurate for some demographics **(1)**.
This raises concerns about discrimination and fairness in automated decision-making **(1)**.

Exam tip: Give a concrete example of bias (facial recognition, recruitment, credit scoring) to demonstrate your understanding.

Worked Example 11 – Discussing privacy

Question: A social media company collects data about users' browsing habits. Discuss the ethical issues this raises. [4 marks]

Model answer:
Privacy concerns:
Users may not be fully aware of how much data is being collected about them **(1)**.
The data could be used in ways the user did not expect or consent to **(1)**.
Consent issues:
Privacy policies are often long and difficult to understand, so consent may not be truly informed **(1)**.

Users may feel forced to accept data collection if they want to use the service **(1)**.

Exam tip: For "discuss" questions on ethics, consider multiple angles — privacy, consent, ownership, and fairness are all relevant perspectives.

Worked Example 12 – Evaluating automation impact

Question: A factory is considering replacing human workers with robots. Evaluate the ethical implications of this decision. [4 marks]

Model answer:
Arguments for automation:
Robots can perform dangerous tasks, improving worker safety **(1)**.
Increased efficiency could make products cheaper for consumers **(1)**.
Arguments against automation:
Workers would lose their jobs and income, causing hardship for them and their families **(1)**.
The company has a responsibility to its employees, and replacement with machines could be seen as unfair **(1)**.

Exam tip: "Evaluate" requires you to consider arguments on both sides. Present a balanced view, even if the question seems to have an obvious answer.

Worked examples summary

This section summarises the 12 worked examples covered in this topic, organised by difficulty level and question type. Use these examples to practise exam technique and understand what examiners are looking for.

Command words for Topic 5

Topic 5 questions often use these command words — understand what each requires:

- **State / Identify:** Give a brief, direct answer — no explanation needed
- **Describe:** Set out characteristics or features — more detail than "state"
- **Explain:** Give reasons and make relationships clear — use "because", "this means"
- **Discuss:** Consider different aspects or viewpoints — show multiple perspectives
- **Evaluate:** Judge worth or importance — weigh up arguments and reach a conclusion

Exam strategy for issues and impact questions

Topic 5 questions often ask you to discuss or evaluate, which requires considering multiple perspectives:

For environmental questions

- Consider all lifecycle stages (manufacture, use, disposal)
- Name specific impacts (e-waste, carbon emissions, resource extraction)
- Link replacement cycles to environmental harm

For IP protection questions

- Match protection type to what is being protected
- Copyright = creative works (automatic)
- Patent = inventions (must apply)
- Trademark = brand identity

For Data Protection questions

- Know the key principles (security, accuracy, purpose limitation)
- Link requirements to specific scenarios
- Consider both organisation obligations and individual rights

For ethical questions

- Present multiple perspectives
- Consider privacy, consent, ownership, fairness
- For AI: think about accountability, bias, safety
- Avoid one-sided answers

Example distribution by section

Example	Topic	Question type	Marks	Key skill
1	Manufacturing impact	Describe	2	Link issue to effect
2	E-waste problems	Explain	2	Name specific hazards
3	Replacement cycles	Explain	2	Chain of reasoning (2024 exam)
4	Data centre energy	Describe	2	Two linked points
5	IP protection (invention)	State	1	Simple recall

Example	Topic	Question type	Marks	Key skill
6	IP matching table	Complete table	2	Match protection types (2024 exam)
7	Copyright benefits	Explain	2	Link protection to benefit
8	Data Protection Act	State	2	Recall key requirements
9	AI accountability	Explain	2	Identify ethical issue
10	AI bias	Explain	3	Extended reasoning
11	Privacy ethics	Discuss	4	Multiple perspectives
12	Automation evaluation	Evaluate	4	Balance arguments

Quick reference

Use these tables for quick revision before exams. The definitions are written in mark-scheme style — memorising these exact phrasings will help you score full marks on definition and identification questions.

> **Revision strategy:** For Topic 5 questions, you need to show awareness of multiple perspectives. Practise linking environmental impacts to the three lifecycle stages, matching IP types to appropriate protection, and presenting balanced arguments for ethical questions.

Key terms and definitions

Term	Definition (mark-scheme style)
E-waste	Electronic waste — discarded electrical devices and components
Replacement cycle	How frequently devices are replaced with newer models
Copyright	Legal protection for original creative works that is automatic
Patent	Legal protection for inventions that must be applied for
Trademark	Legal protection for brand identity (logos, names, slogans)
Licensing	Legal permission to use protected work under specific terms
Data Protection Act	UK law controlling how personal data is collected, stored, and used
Personal data	Any information that can identify a living individual
Privacy	The right to control information about oneself

Term	Definition (mark-scheme style)
Consent	Informed permission given for data collection or use
Accountability	Responsibility for actions and their consequences
Bias (in AI)	Unfair prejudice in AI systems resulting from biased training data

IP protection quick match

What is being protected?	Protection method
Song / music	Copyright
Photograph / image	Copyright
Software / program code	Copyright
Written text / book	Copyright
Hardware invention	Patent
New manufacturing process	Patent
Company logo	Trademark
Advertising slogan	Trademark
Product name	Trademark

Environmental impact summary

Lifecycle stage	Key impacts	Exam phrases
Manufacture	Resource extraction, factory energy, water use	"Extraction of metals", "fossil fuels", "water consumption"
Use	Electricity consumption, data centre energy	"Carbon footprint", "electricity from fossil fuels"
Disposal	E-waste, toxic materials, landfill pollution	"Lead/mercury", "hazardous materials", "landfill"
Short replacement cycles	Increases all of the above	"More manufacturing", "more disposal", "more e-waste"

Ethical issues with AI summary

Issue	Key question	Example
Accountability	Who is responsible when AI makes mistakes?	Self-driving car accident
Bias	Does AI treat all groups fairly?	Facial recognition accuracy
Safety	Will AI systems operate reliably?	Medical diagnosis AI
Job displacement	What happens to replaced workers?	Factory automation
Privacy	Is data collection acceptable?	Facial recognition in public

Typical pitfalls

These are common mistakes students make in Topic 5 questions. Avoiding these errors will help you gain marks in the exam. Many of these pitfalls involve being too vague or not addressing all parts of the question.

Pitfall 1 — Confusing copyright and patent.
Wrong: "Software is protected by a patent."
Right: Software code is protected by **copyright** (creative work). **Patents** protect inventions, typically hardware.

Pitfall 2 — Forgetting patents must be applied for.
Wrong: "The inventor automatically has a patent."
Right: Copyright is automatic, but patents **must be applied for** and granted by a patent office.

Pitfall 3 — Only considering one lifecycle stage.
Wrong: "The environmental impact is just about e-waste."
Right: Consider all three stages: **manufacture** (resource extraction), **use** (energy consumption), and **disposal** (e-waste).

Pitfall 4 — Vague environmental answers.
Wrong: "It's bad for the environment."
Right: Be specific: "Manufacturing requires extraction of rare earth metals, which damages the environment through mining."

Pitfall 5 — Mixing up ethical and legal.
Wrong: Using "illegal" and "unethical" interchangeably.
Right: Legal = required/prohibited by law. **Ethical** = questions of right and wrong. Something can be legal but unethical, or illegal but arguably ethical.

Pitfall 6 — One-sided ethical answers.
Wrong: Only presenting one perspective on an ethical issue.
Right: For "discuss" or "evaluate" questions, present **multiple viewpoints** — arguments for and against.

Pitfall 7 — Forgetting to link replacement cycles.
Wrong: "Short replacement cycles are bad."
Right: Explain **why**: "Shorter replacement cycles lead to more manufacturing and more disposal, increasing environmental impact."

Pitfall 8 — Confusing trademark and copyright.
Wrong: "A logo is protected by copyright."
Right: A logo is **brand identity**, protected by **trademark**. Copyright protects creative works like music and text.

Pitfall 9 — Generic Data Protection answers.
Wrong: "They must protect the data."
Right: Be specific about **which** requirement: "Keep data secure", "Only use data for its stated purpose", "Keep data accurate".

Pitfall 10 — Missing the question about AI accountability.
Wrong: "AI is dangerous because it might make mistakes."
Right: Focus on **accountability**: "When AI makes mistakes, it is unclear who is responsible — the developer, manufacturer, or user."

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